NEUROFUSION AI | PRELIMINARY AI ANALYSIS REPORT

REPORT ID: REP-20261102-849488

Automated Multi-Modal AI Results • Stage 1: Pending Clinical Sign-Off

November 02, 2026 • 16:42 UTC

PATIENT INFORMATION	ATTENDING NEUROLOGIST	EXAMINATION DETAILS
ID: PAT-1083 **Name:** Eleanor Vance **Age/Gender:** 67 yrs / Female **Status:** Parkinson's Disease (Hoehn & Yahr Stage II)	**Name:** Dr. Clara Rodriguez, MD **License:** MD-NEURO-77491 **Dept:** Neurology & Movement Disorders **Hospital:** Metropolitan Neurological Inst.	**Scan Date:** 2026-11-02 **Report Date:** 2026-11-02 **Modality:** Mri **Device/Protocol:** NeuroFusionAI Protocol PARKINSON v2.0

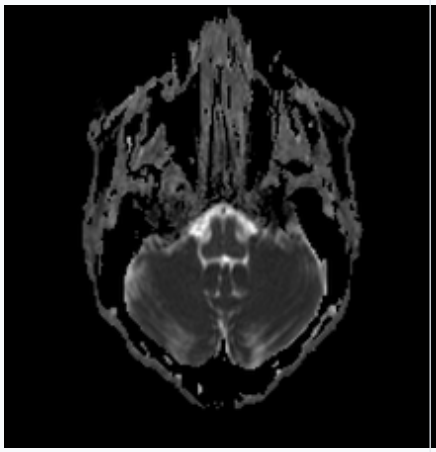
PRIMARY AI PREDICTION: NORMAL

Confidence: 100.0%
Target: Parkinson's Disease

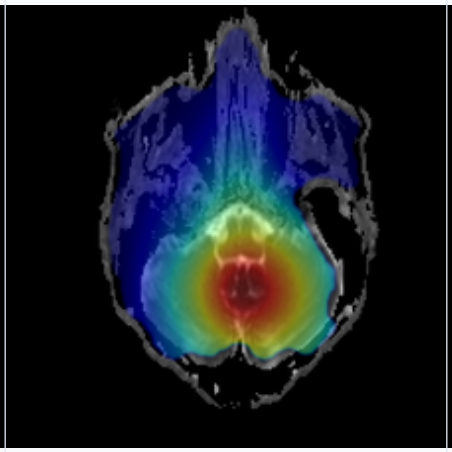
Class Probability Distribution: Normal: 100.0% • Parkinson: 0.1%

AI Confidence Calibrated via Temperature Scaling

1. MRI NEUROIMAGING & CONVOLUTIONAL GRAD-CAM EXPLAINABILITY



Input MRI Scan (Axial)



Grad-CAM Activation Overlay

Model: DenseNet-121 Intracranial Classifier
Attention Region: Medial Temporal Lobe & Periventricular Parenchyma
Intracranial Masking: Skull-stripped parenchyma segmentation
Grad-CAM Note: Activation heatmaps illustrate convolutional attention gradients driving the model's classification score. Not direct histological pathology.

XAI Interpretation

7. CLINICAL DECISION SUPPORT SYSTEM (CDSS) RECOMMENDATIONS

RISK LEVEL:
LOW CLINICAL RISK

Urgency: Routine Outpatient Follow-up

Evidence-Based Diagnostic Recommendations:
• Correlate AI automated assessment with comprehensive MDS-UPDRS motor and cognitive exam.
• Recommend DaTscan SPECT dopamine transporter neuroimaging to confirm nigrostriatal deficit.
• Order formal neuropsychological battery (MoCA / MMSE) to track longitudinal cognitive trajectory.
• Initiate outpatient physical therapy and occupational therapy motor rehabilitation.

PATIENT LONGITUDINAL DIAGNOSTIC HISTORY

Date	Modality	Prediction	Confidence	Clinical Status
2026-11-02	Mri	Normal	100.0%	Verified Match
2026-11-02	Fusion	Normal	95.8%	Verified Match

2026-11-02	Voice	Parkinson	89.5%	Verified Match
2026-11-02	Spiral	Parkinson	87.4%	Verified Match

PRELIMINARY AI SYSTEM OBSERVATIONS:	CLINICAL APPROVAL STATUS
All required automated analytical pipelines (CNN/ViT, EEG feature extraction, XAI, and CDSS) have successfully executed. Final clinical diagnosis and treatment decisions await attending physician review.	*Preliminary AI analysis for decision support. Official physician certification pending.* Status: **AWAITING PHYSICIAN REVIEW & APPROVAL**

REGULATORY MEDICAL DISCLAIMER & CLINICAL INTENDED USE:

NeuroFusionAI is an investigational Artificial Intelligence Clinical Decision Support System (CDSS) designed to augment neurological assessment. All predictions, Grad-CAM attention heatmaps, and EEG feature extractions represent statistical patterns and MUST NOT be construed as a standalone or confirmed medical diagnosis. Clinical treatment decisions must be made solely by a board-certified physician in conjunction with the patient's full medical history.